Breathing and Exchange of Gases

14.0 Why We Breathe - The Chapter's Starting Point

Oxygen (O_2) is utilised by organisms to **indirectly** break down simple molecules like **glucose, amino acids and fatty acids** to derive energy for various activities. **Carbon dioxide (CO_2)**, which is **harmful**, is released during these **catabolic reactions**. It therefore becomes essential that O_2 is **continuously provided** to the cells and the CO_2 produced by the cells is **released out**.

- **Breathing:** the process of exchange of O_2 from the atmosphere with CO_2 produced by the cells. It is commonly known as **respiration**.

14.1 Respiratory Organs

Mechanisms of breathing **vary among different groups of animals** depending mainly on their **habitats** and **levels of organisation**.

Animal group	Respiratory structure / mechanism	Named term
Lower invertebrates - sponges, coelenterates, flatworms	Exchange O_2 with CO_2 by **simple diffusion over their entire body surface**	Diffusion
Earthworms	Use their **moist cuticle** to exchange gases	Cutaneous
Insects	A network of tubes (**tracheal tubes**) transports atmospheric air within the body	Tracheal
Most aquatic arthropods and molluscs	Special **vascularised structures called gills**	**Branchial respiration**
Terrestrial forms	**Vascularised bags called lungs**	**Pulmonary respiration**

Among **vertebrates**, **fishes** use gills whereas **amphibians, reptiles, birds and mammals** respire through **lungs**. Amphibians like **frogs** can **also** respire through their **moist skin** (**cutaneous respiration**).

14.1.1 Human Respiratory System

We have a pair of **external nostrils** opening out **above the upper lips**. These lead to a **nasal chamber** through the **nasal passage**. The nasal chamber opens into the **pharynx**, a portion of which is the **common passage for food and air**. The pharynx opens through the **larynx** region into the **trachea**.

- **Larynx:** a **cartilaginous box** which helps in **sound production** and hence is called the **sound box**.

During **swallowing**, the **glottis** can be covered by a **thin elastic cartilaginous flap called epiglottis** to **prevent the entry of food into the larynx**.

14.1.1a The Airway, Branch by Branch

1. **Trachea -** a **straight tube** extending up to the **mid-thoracic cavity**.
2. The trachea **divides at the level of the 5th thoracic vertebra** into a **right and left primary bronchi**. (Each such division gives a **Bronchus**.)
3. Each **bronchus** undergoes **repeated divisions** to form the **secondary and tertiary bronchi and bronchioles**, ending up in **very thin terminal bronchioles**. (A single such branch is a **Bronchiole**.)
4. Each **terminal bronchiole** gives rise to a number of **very thin, irregular-walled and vascularised bag-like structures called alveoli**.

The **trachea, primary, secondary and tertiary bronchi, and initial bronchioles** are supported by **incomplete cartilaginous rings**. The **branching network of bronchi, bronchioles and alveoli comprises the lungs -** we have **two lungs**. Each **Lung** is covered by a **double-layered pleura**, with **pleural fluid** between its two layers, and this **pleural fluid reduces friction on the lung-surface**. Of the two **pleural membranes**, the **outer** one is in **close contact with the thoracic lining** whereas the **inner** one is in contact with the **lung surface**.

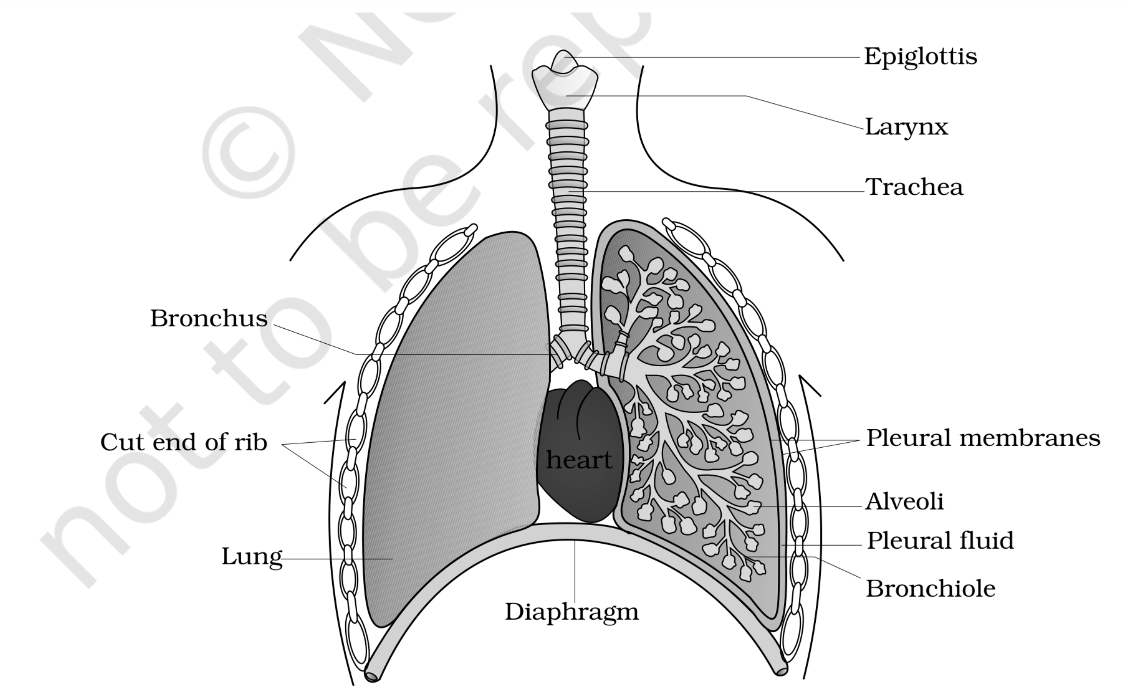

*Fig. 14.1 - Diagrammatic view of the human respiratory system (a sectional view of the left lung is also shown). Labelled from top down: **Epiglottis**, **Larynx**, **Trachea**, and the **Bronchus** entering each lung, dividing into a **Bronchiole** that ends in **Alveoli**; the **Cut end of rib** and the two **Pleural membranes** with **Pleural fluid** between them enclose the **Lung**, with the **heart** lying between the lungs and the dome-shaped **Diaphragm** below.*

14.1.1b Conducting Part vs Exchange Part

Part	Extent	Function
Conducting part	From the **external nostrils up to the terminal bronchioles**	Transports the atmospheric air to the alveoli, **clears it from foreign particles**, **humidifies it** and **brings the air to body temperature**
Respiratory or exchange part	The **alveoli and their ducts**	The **site of actual diffusion** of O_2 and CO_2 between **blood and atmospheric air**

14.1.1c The Thoracic Chamber

The lungs are situated in the **thoracic chamber**, which is anatomically an **air-tight chamber**. It is formed **dorsally by the vertebral column**, **ventrally by the sternum**, **laterally by the ribs** and on the **lower side by the dome-shaped diaphragm**. The anatomical setup is such that **any change in the volume of the thoracic cavity is reflected in the pulmonary (lung) cavity**. Such an arrangement is **essential for breathing**, as **we cannot directly alter the pulmonary volume**.

> *[NOTE] Respiration in humans involves the following **five steps**, in this order: (i) **Breathing or pulmonary ventilation**, by which atmospheric air is **drawn in** and **CO_2-rich alveolar air is released out**; (ii) **diffusion of gases** (O_2 and CO_2) **across the alveolar membrane**; (iii) **transport of gases by the blood**; (iv) **diffusion of O_2 and CO_2 between blood and tissues**; and (v) **utilisation of O_2 by the cells for catabolic reactions and resultant release of CO_2 (cellular respiration)**.*

14.2 Mechanism of Breathing

Breathing involves **two stages**: **inspiration**, during which **atmospheric air is drawn in**, and **expiration**, by which the **alveolar air is released out**. The movement of air into and out of the lungs is carried out by creating a **pressure gradient** between the lungs and the atmosphere.

- **Inspiration** can occur if the pressure within the lungs (**intra-pulmonary pressure**) is **less than the atmospheric pressure**, i.e. there is a **negative pressure in the lungs with respect to atmospheric pressure**.
- **Expiration** takes place when the **intra-pulmonary pressure is higher than the atmospheric pressure**.

The **diaphragm** and a specialised set of muscles **- external and internal intercostals** between the ribs **-** help in the generation of such pressure gradients.

14.2a **Inspiration, Step by Step**

1. Inspiration is initiated by the **contraction of the diaphragm**, which **increases the volume of the thoracic chamber in the antero-posterior axis**. In the figure this is the **Diaphragm contracted** state.

2. **Contraction of the external inter-costal muscles lifts up the ribs and the sternum - Ribs and sternum raised -** causing an **increase in the volume of the thoracic chamber in the dorso-ventral axis**, expanding the **Rib cage**.

3. The overall **increase in thoracic volume** causes a **similar increase in pulmonary volume**, so the **Volume of thorax increased**.

4. An increase in pulmonary volume **decreases the intra-pulmonary pressure to less than the atmospheric pressure**, which **forces air from outside to move into the lungs - Air entering lungs**. This is **inspiration**.

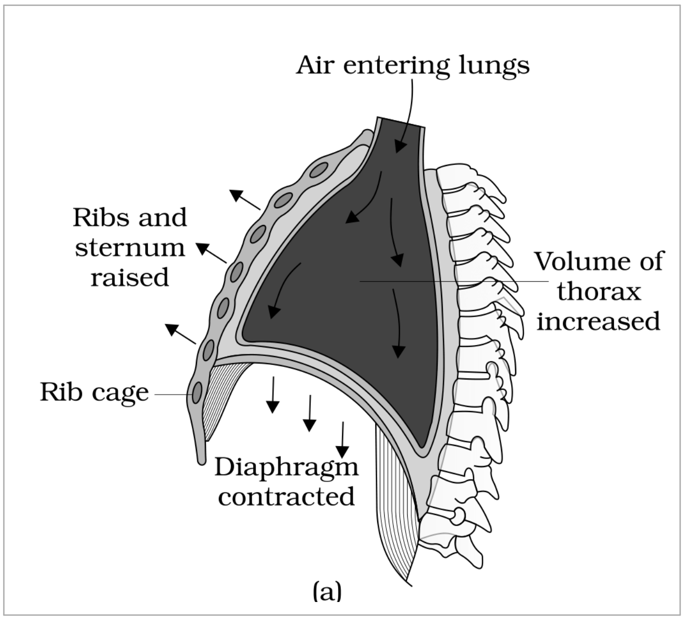

*Fig. 14.2 (a) - Mechanism of breathing showing **inspiration**: the **Rib cage** with **Ribs and sternum raised**, the **Diaphragm contracted** and flattened, the **Volume of thorax increased**, and **Air entering lungs**.*

14.2b **Expiration, Step by Step**

1. **Relaxation of the diaphragm and the inter-costal muscles** returns the **diaphragm and sternum to their normal positions -** the **Diaphragm relaxed and arched upwards**, and the **Ribs and sternum returned to original position - reducing the thoracic volume and thereby the pulmonary volume**, so the **Volume of thorax decreased**.

2. This reduction leads to an **increase in intra-pulmonary pressure to slightly above the atmospheric pressure**, causing the **expulsion of air from the lungs - Air expelled from lungs**. This is **expiration**.

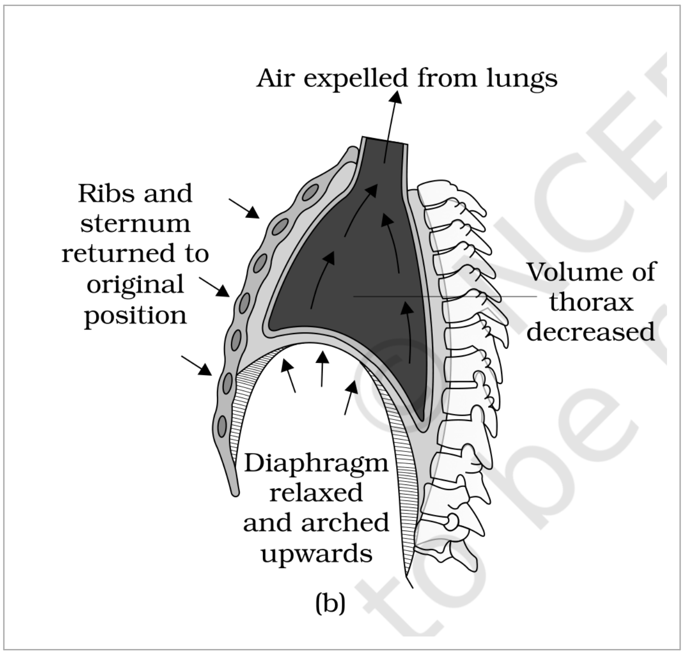

*Fig. 14.2 (b) - Mechanism of breathing showing **expiration**: the **Ribs and sternum returned to original position**, the **Diaphragm relaxed and arched upwards**, the **Volume of thorax decreased**, and **Air expelled from lungs**.*

We can **increase the strength of inspiration and expiration** with the help of **additional muscles in the abdomen**. On an average, a **healthy human breathes 12-16 times per minute**.

- **Spirometer:** the instrument with which the **volume of air involved in breathing movements** can be estimated. It helps in the **clinical assessment of pulmonary functions**.

14.2.1 Respiratory Volumes and Capacities

A **healthy man can inspire or expire approximately 6000 to 8000 mL of air per minute**. The individual volumes below are what a spirometer measures.

Volume	Definition	Average value
Tidal Volume (TV)	Volume of air **inspired or expired during a normal respiration**	Approx. **500 mL**
Inspiratory Reserve Volume (IRV)	**Additional volume of air a person can inspire by a forcible inspiration**	**2500 mL to 3000 mL**
Expiratory Reserve Volume (ERV)	**Additional volume of air a person can expire by a forcible expiration**	**1000 mL to 1100 mL**
Residual Volume (RV)	Volume of air **remaining in the lungs even after a forcible expiration**	**1100 mL to 1200 mL**

By **adding up a few respiratory volumes**, one can derive various **pulmonary capacities**, which **can be used in clinical diagnosis**.

Capacity	Definition	Formula
Inspiratory Capacity (IC)	**Total volume of air a person can inspire after a normal expiration**	**TV + IRV**
Expiratory Capacity (EC)	**Total volume of air a person can expire after a normal inspiration**	**TV + ERV**
Functional Residual Capacity (FRC)	**Volume of air that will remain in the lungs after a normal expiration**	**ERV + RV**
Vital Capacity (VC)	The **maximum volume of air a person can breathe in after a forced expiration**, or **breathe out after a forced inspiration**	**ERV + TV + IRV**
Total Lung Capacity (TLC)	**Total volume of air accommodated in the lungs at the end of a forced inspiration**	**RV + ERV + TV + IRV**, i.e. **vital capacity + residual volume**

☆ ***[MEMORY AID - not in NCERT]** The four **capacities built by addition** all start from a volume you already know: **I**nspiratory **C**apacity adds the inspiratory pair (TV + IRV), **E**xpiratory **C**apacity adds the expiratory pair (TV + ERV) **-** the letter before 'C' tells you which reserve joins the tidal volume. **FRC** is what is left behind (ERV + RV), and **TLC** is simply **VC + RV**.*

14.3 Exchange of Gases

Alveoli are the primary sites of exchange of gases. Exchange of gases **also** occurs **between blood and tissues**. O_2 and CO_2 are exchanged at these sites by **simple diffusion**, mainly based on **pressure/concentration gradient**. **Solubility of the gases** and the **thickness of the membranes** involved in diffusion are **important factors that can affect the rate of diffusion**.

- **Partial pressure:** the pressure contributed by an **individual gas in a mixture of gases**. It is represented as **pO_2** for oxygen and **pCO_2** for carbon dioxide.

Table 14.1 - Partial pressures (in mm Hg) of oxygen and carbon dioxide at different parts involved in diffusion

Respiratory gas	Atmospheric air	Alveoli	Blood (deoxygenated)	Blood (oxygenated)	Tissues
pO_2	**159**	**104**	**40**	**95**	**40**
pCO_2	**0.3**	**40**	**45**	**40**	**45**

The data in Table 14.1 indicate a **concentration gradient for oxygen from alveoli to blood** and **from blood to tissues**. Similarly, a **gradient for CO_2 is present in the opposite direction**, i.e. **from tissues to blood** and **from blood to alveoli**.

As the **solubility of CO_2 is 20-25 times higher than that of O_2**, the **amount of CO_2 that can diffuse through the diffusion membrane per unit difference in partial pressure is much higher** compared to that of O_2.

14.3a The Diffusion Membrane

The **diffusion membrane** is made up of **three major layers**: the **thin squamous epithelium of alveoli**, the **endothelium of alveolar capillaries** and the **basement substance** in between them. The **basement substance** is composed of a **thin basement membrane** supporting the squamous epithelium and the **basement membrane surrounding the single-layer endothelial cells of the capillaries**. However, its **total thickness is much less than a millimetre**.

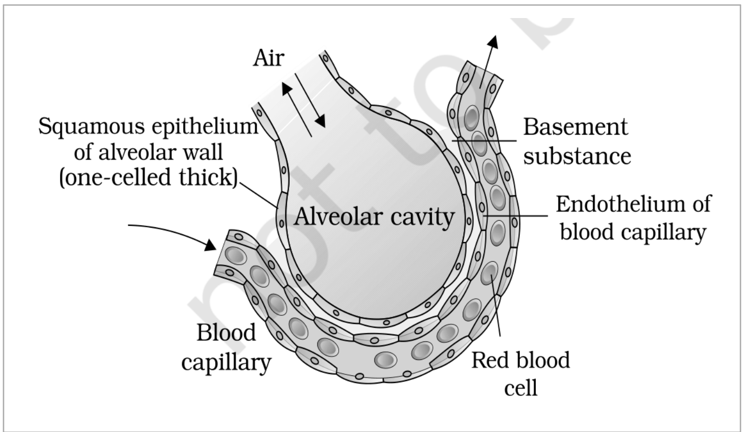

*Fig. 14.4 - A diagram of a section of an **alveolus** with a **pulmonary capillary**. The **Air** in the **Alveolar cavity** is separated from the **Blood capillary** by the three layers of the diffusion membrane: the **Squamous epithelium of alveolar wall**, the **Basement substance** and the **Endothelium of blood capillary**; a **Red blood cell** is shown inside the capillary.*

ⓘ ***[NOTE] All the factors in our body are favourable for diffusion of O_2 from alveoli to tissues and of CO_2 from tissues to alveoli**. This is why **diffusion of gases occurs in the alveolar region only and not in the other parts of the respiratory system**: only there does a **vascularised, thin, three-layered diffusion membrane** lie between air and blood, while the rest of the tract **-** the **conducting part -** has thick, cartilage-supported walls and merely carries, cleans, humidifies and warms the air.*

14.4 Transport of Gases

Blood is the medium of transport for O_2 and CO_2.

Gas	Carried by	Share
O_2	**RBCs** in the blood	About **97 per cent**
O_2	**Dissolved state through the plasma**	Remaining **3 per cent**
CO_2	**RBCs**	Nearly **20-25 per cent**
CO_2	As **bicarbonate**	About **70 per cent**
CO_2	**Dissolved state through plasma**	About **7 per cent**

14.4.1 Transport of Oxygen

- **Haemoglobin:** a **red-coloured, iron-containing pigment** present in the RBCs.

O_2 can **bind with haemoglobin in a reversible manner** to form **oxyhaemoglobin**. **Each haemoglobin molecule can carry a maximum of four molecules of O_2**. Binding of oxygen with haemoglobin is **primarily related to the partial pressure of O_2**. **Partial pressure of CO_2, hydrogen ion concentration and temperature** are the **other factors which can interfere with this binding**.

- **Oxygen dissociation curve:** the **sigmoid curve** obtained when the **percentage saturation of haemoglobin with O_2** is plotted against **pO_2**. It is **highly useful in studying the effect of factors like pCO_2 and H^+ concentration on the binding of O_2 with haemoglobin**.

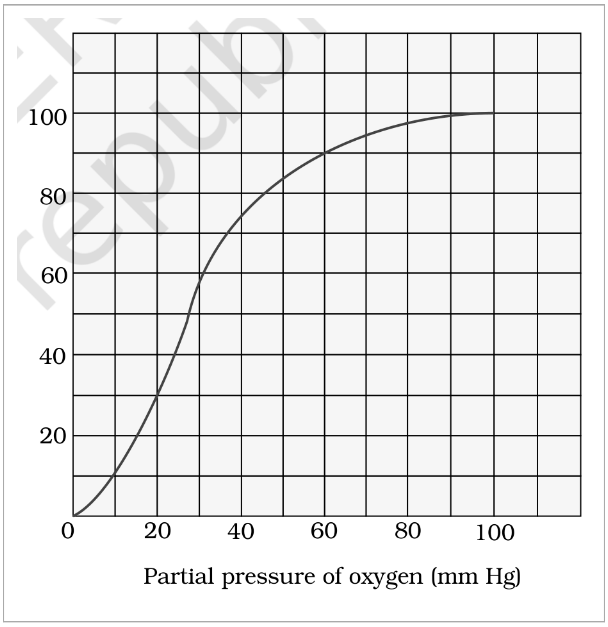

*Fig. 14.5 - **Oxygen dissociation curve**. The vertical axis is the **Percentage saturation of haemoglobin with oxygen** and the horizontal axis is the **Partial pressure of oxygen (mm Hg)**; the plotted sigmoid line is the **Oxygen dissociation curve** itself.*

Site	Conditions	Result
In the alveoli	**High pO_2, low pCO_2, lesser H^+ concentration** and **lower temperature**	The factors are **all favourable for the formation of oxyhaemoglobin**
In the tissues	**Low pO_2, high pCO_2, high H^+ concentration** and **higher temperature**	Conditions are **favourable for dissociation of oxygen from oxyhaemoglobin**

This clearly indicates that **O_2 gets bound to haemoglobin at the lung surface** and **gets dissociated at the tissues**. Under **normal physiological conditions**, **every 100 mL of oxygenated blood can deliver around 5 mL of O_2 to the tissues**.

> ⓘ ***[NOTE] Effect of pCO_2 on oxygen transport (Exercise 8):*** *a **high pCO_2**, which also raises the H^+ concentration, is one of the tissue-side conditions that **favours dissociation of O_2 from oxyhaemoglobin**. A **low pCO_2**, as in the alveoli, favours **formation** of oxyhaemoglobin instead. So pCO_2 does not carry the oxygen, but it decides **where** the oxygen is released.*

14.4.2 Transport of Carbon dioxide

CO_2 is carried by **haemoglobin as carbamino-haemoglobin** (about **20-25 per cent**). This binding is **related to the partial pressure of CO_2. pO_2 is a major factor which could affect this binding**.

- When **pCO_2 is high** and **pO_2 is low**, as in the **tissues**, **more binding of carbon dioxide occurs**.
- When **pCO_2 is low** and **pO_2 is high**, as in the **alveoli**, **dissociation of CO_2 from carbamino-haemoglobin takes place -** i.e. the CO_2 bound to haemoglobin from the tissues is **delivered at the alveoli**.

- **Carbonic anhydrase:** RBCs contain a **very high concentration** of this enzyme, and **minute quantities of the same is present in the plasma too**.

This enzyme facilitates the following reaction **in both directions**:

$$CO_2 + H_2O \xleftrightarrow{\text{carbonic anhydrase}} H_2CO_3 \xleftrightarrow{\text{carbonic anhydrase}} HCO_3^- + H^+$$

1. **At the tissue site**, where **pCO_2 is high due to catabolism**, CO_2 **diffuses into blood (RBCs and plasma)** and forms **HCO_3^- and H^+**.
2. **At the alveolar site**, where **pCO_2 is low**, the reaction **proceeds in the opposite direction**, leading to the formation of **CO_2 and H_2O**.
3. Thus CO_2 **trapped as bicarbonate at the tissue level and transported to the alveoli is released out as CO_2**.

Every 100 mL of deoxygenated blood delivers approximately 4 mL of CO_2 to the alveoli.

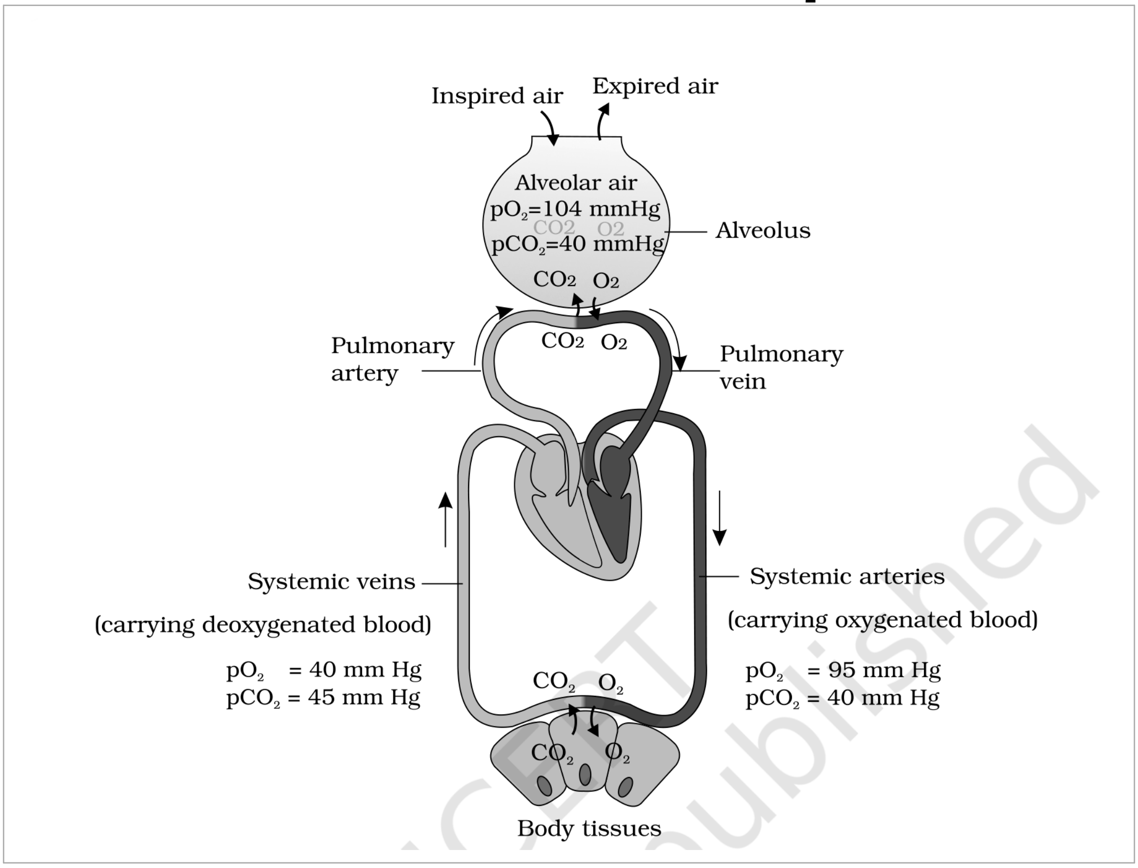

*Fig. 14.3 - Diagrammatic representation of exchange of gases at the **Alveolus** and the **Body tissues** with blood, and transport of oxygen and carbon dioxide. **Inspired air** enters and **Expired air** leaves the alveolus, whose **Alveolar air** exchanges **CO_2** and **O_2** with the blood: the **Pulmonary artery** brings deoxygenated blood to the alveolus and the **Pulmonary vein** carries oxygenated blood away, while the **Systemic arteries** deliver it to the **Body tissues** and the **Systemic veins** return it. In the figure the two gases are marked **CO2** and **O2**.*

14.5 Regulation of Respiration

Human beings have a **significant ability to maintain and moderate the respiratory rhythm** to **suit the demands of the body tissues**. This is done by the **neural system**.

- **Respiratory rhythm centre:** a **specialised centre present in the medulla region of the brain**, **primarily responsible** for this regulation.
- **Pneumotaxic centre:** another centre present in the **pons region** of the brain which **can moderate the functions of the respiratory rhythm centre**. **Neural signal from this centre can reduce the duration of inspiration** and thereby **alter the respiratory rate**.
- **Chemosensitive area:** situated **adjacent to the rhythm centre**, **highly sensitive to CO_2 and hydrogen ions**. **Increase in these substances can activate this centre**, which in turn **can signal the rhythm centre to make necessary adjustments in the respiratory process by which these substances can be eliminated**.
- **Receptors associated with the aortic arch and carotid artery also** can **recognise changes in CO_2 and H^+ concentration** and **send necessary signals to the rhythm centre for remedial actions**.

> ⓘ ***[NOTE] The role of oxygen in the regulation of respiratory rhythm is quite insignificant.*** *The regulation is driven by **CO_2 and H^+**, not by O_2* ***-*** *a favourite NEET assertion-reason trap.*

14.6 Disorders of Respiratory System

Disorder	What happens	Cause / note
Asthma	A **difficulty in breathing causing wheezing**	Due to **inflammation of bronchi and bronchioles**
Emphysema	A **chronic disorder** in which **alveolar walls are damaged**, due to which the **respiratory surface is decreased**	**One of the major causes of this is cigarette smoking**
Occupational Respiratory Disorders	In **certain industries**, **especially those involving grinding or stone-breaking**, **so much dust is produced that the defence mechanism of the body cannot fully cope with the situation**	**Long exposure** can give rise to **inflammation** leading to **fibrosis** (**proliferation of fibrous tissues**) and thus causing **serious lung damage**. **Workers in such industries should wear protective masks.**

Recap Quick Recap

- Cells **utilise oxygen for metabolism and produce energy** along with substances like **carbon dioxide which is harmful**. Animals have **evolved different mechanisms** for the **transport of oxygen to the cells** and for the **removal of carbon dioxide** from there. We have a **well developed respiratory system** comprising **two lungs and associated air passages**.
- The **first step in respiration is breathing**, by which atmospheric air is **taken in (inspiration)** and the alveolar air is **released out (expiration)**. **Exchange of O_2 and CO_2 between deoxygenated blood and alveoli**, **transport of these gases throughout the body by blood**, **exchange of O_2 and CO_2 between the oxygenated blood and tissues** and **utilisation of O_2 by the cells (cellular respiration)** are the **other steps** involved.
- **Inspiration and expiration** are carried out by **creating pressure gradients between the atmosphere and the alveoli** with the help of **specialised muscles - intercostals and diaphragm**. **Volumes of air** involved in these activities **can be estimated with the help of a spirometer** and are **of clinical significance**.
- **Exchange of O_2 and CO_2 at the alveoli and tissues occurs by diffusion.** The **rate of diffusion is dependent on the partial pressure gradients of O_2 (pO_2) and CO_2 (pCO_2)**, **their solubility** as well as the **thickness of the diffusion surface**. These factors in our body **facilitate diffusion of O_2 from the alveoli to the deoxygenated blood as well as from the oxygenated blood to the tissues**, and are **favourable for the diffusion of CO_2 in the opposite direction**, i.e. **from tissues to alveoli**.
- **Oxygen is transported mainly as oxyhaemoglobin.** In the **alveoli**, where pO_2 is **higher**, O_2 **gets bound to haemoglobin**, which is **easily dissociated at the tissues** where pO_2 is **low** and pCO_2 and H^+ concentration are **high**.
- **Nearly 70 per cent of carbon dioxide is transported as bicarbonate (HCO_3^-)** with the help of the enzyme **carbonic anhydrase**. **20-25 per cent of carbon dioxide is carried by haemoglobin as carbamino-haemoglobin**: in the **tissues**, where pCO_2 is **high**, it **gets bound to blood**, whereas in the **alveoli**, where pCO_2 is **low** and pO_2 is **high**, it **gets removed from the blood**.
- **Respiratory rhythm is maintained by the respiratory centre in the medulla region of the brain.** A **pneumotaxic centre in the pons region** of the brain and a **chemosensitive area in the medulla can alter the respiratory mechanism**.

Terms Used in the Exercises

Ex 9 A man going up a hill (high altitude)

Going up a hill means **climbing into thinner air**, so the **partial pressure of oxygen (pO_2) in the atmospheric air falls below its sea-level value of 159 mm Hg**. Because **binding of oxygen with haemoglobin is primarily related to the partial pressure of O_2**, a lower atmospheric pO_2 means a lower alveolar pO_2, so **less oxyhaemoglobin is formed** and **less O_2 is delivered per 100 mL of blood** than the usual 5 mL. The body compensates through the very mechanism of SS14.5: the **respiratory rhythm centre**, prompted by the **chemosensitive area** and the **receptors of the aortic arch and carotid artery**, **increases the rate and depth of breathing -** the climber breathes **faster and deeper** than the normal **12-16 times per minute**, using the **additional muscles in the abdomen** to strengthen inspiration and expiration.

Ex 11 Why the oxygen dissociation curve is sigmoidal

Each haemoglobin molecule can carry a maximum of four molecules of O_2, and binding is **reversible**. The **four binding events are not independent**: the first O_2 molecule binds with difficulty, but once bound it makes the **remaining sites bind more readily**, and the last site fills only when pO_2 is high. So saturation rises **slowly at low pO_2**, **steeply at intermediate pO_2**, and **flattens as the four sites fill up** - which is exactly the **S-shaped (sigmoid) curve** of Fig. 14.5. The **steep middle portion** is the physiologically useful part: it is why a **small fall in pO_2 at the tissues** (40 mm Hg against the alveolar 104 mm Hg) **unloads a large amount of O_2**.

Ex 12 Hypoxia

Hypoxia is a **deficiency of oxygen reaching the body tissues -** the tissues receive **less O_2 than they need** for the catabolic reactions that release energy. Reading it against this chapter, hypoxia is what results whenever any link in the five-step chain weakens: **reduced ventilation** (as in **asthma**, where inflammation of bronchi and bronchioles obstructs breathing), a **reduced respiratory surface** (as in **emphysema**, where alveolar walls are damaged, or in the **fibrosis** of occupational respiratory disorders), a **fall in atmospheric pO_2** (high altitude), or **too little haemoglobin** to carry the **97 per cent** of O_2 that travels in the RBCs.

Ex 14 Tidal volume in an hour

Tidal Volume (TV) is the **volume of air inspired or expired during a normal respiration**, approximately **500 mL**. A **healthy human breathes 12-16 times per minute**, i.e. **720 to 960 breaths per hour**. Multiplying by the tidal volume gives **about 360,000 to 480,000 mL per hour**, i.e. roughly **360 to 480 litres of air per hour**. (This agrees with the chapter's own figure that a healthy man can inspire or expire approximately **6000 to 8000 mL of air per minute**.)

> ☆ ***[MEMORY AID - not in NCERT]*** *For the MCQ in Exercise 5, read Table 14.1 across one row at a time.* ***Atmospheric air vs alveolar air****: oxygen falls* ***159 to 104*** *(so atmospheric* ***pO_2 is higher****) while carbon dioxide rises* ***0.3 to 40*** *(so atmospheric* ***pCO_2 is lesser****). Air arriving from outside is* ***oxygen-rich and carbon-dioxide-poor -*** *which is the whole point of ventilating the alveoli.*

Every fact, number, name, qualifier, table row, figure and figure label in NCERT Class 11 Chapter 14 is carried above. Nothing outside the source chapter has been added, except the clearly marked MEMORY AID boxes and the exercise-gap explanations, which are derived only from chapter content.